KCS

KCS stands for Knowledge-Centered Service, a methodology that aims to improve service delivery and knowledge management in service organizations . Some of the main features and benefits of KCS are:

- It integrates the creation and maintenance of knowledge articles with the resolution of customer issues, making knowledge a by-product of service delivery.
- It empowers service agents to capture, structure, reuse, and improve knowledge articles based on their interactions with customers and their feedback.
- It enables service organizations to leverage their collective experience and expertise, reduce costs, increase efficiency, and improve customer satisfaction .
- It follows a set of principles, practices, and techniques that are based on the KCS methodology, such as the KCS Loop, the KCS Solve and Evolve phases, the KCS roles and competencies, and the KCS adoption model.

KCS is not to be confused with Kansas City Southern, a railroad company that operates in the United States and Mexico .

Unit 1 - The cell concept, structure of prokaryotic, eukaryotic cells, plant cells and animal cells, Structure and function of cell membrane, cell organelles and their function, Basics of cell signaling, Structure and use of compound microscope, Basic chemical constituents of living body

- The cell concept is the idea that all living organisms are composed of one or more cells, and that cells are the basic units of structure and function in life.
- Prokaryotic cells are cells that lack a nucleus and other membrane-bound organelles. They are usually unicellular and have a simple structure. Examples of prokaryotes are bacteria and archaea.
- Eukaryotic cells are cells that have a nucleus and other membrane-bound organelles. They are usually multicellular and have a complex structure. Examples of eukaryotes are animals, plants, fungi and protists.
- Plant cells are eukaryotic cells that have some specific features, such as a cell wall, a large central vacuole, chloroplasts and plasmodesmata. Plant cells are involved in photosynthesis, storage, support and communication.
- Animal cells are eukaryotic cells that lack a cell wall, a large central vacuole, chloroplasts and plasmodesmata. Animal cells are involved in movement, digestion, respiration, reproduction and communication.
- The cell membrane is a thin, flexible layer that surrounds the cell and

regulates the movement of substances in and out of the cell. It is composed of a phospholipid bilayer with embedded proteins, carbohydrates and cholesterol. The cell membrane has various functions, such as protection, transport, recognition, signaling and attachment.

- Cell organelles are specialized structures within the cell that perform specific functions. Some examples of cell organelles are:
  - Nucleus: The control center of the cell that contains the genetic material (DNA) and regulates gene expression.
  - Mitochondria: The powerhouses of the cell that produce energy (ATP) by cellular respiration.
  - Ribosomes: The sites of protein synthesis that translate mRNA into amino acids.
  - Endoplasmic reticulum: A network of membranes that transport and modify proteins and lipids.
  - Golgi apparatus: A stack of flattened membranes that sort and package proteins and lipids for secretion or delivery to other organelles.
  - Lysosomes: The digestive system of the cell that break down macromolecules and recycle cellular components.
  - Cytoskeleton: A network of protein filaments that provide shape, support and movement to the cell and its organelles.
  - Cilia and flagella: Hair-like or whip-like structures that extend from the cell surface and enable locomotion or movement of fluids.
- Cell signaling is the process by which cells communicate with each other or with the environment by using chemical or electrical signals. Cell signaling can be classified into four types:
  - Autocrine signaling: When a cell produces and responds to its own signal.
  - Paracrine signaling: When a cell produces a signal that affects nearby cells.
  - Endocrine signaling: When a cell produces a signal that travels through the bloodstream and affects distant cells.
  - Synaptic signaling: When a nerve cell produces a signal that crosses a synapse and affects another nerve cell or a target cell.
- A compound microscope is a device that uses two or more lenses to magnify an object. A compound microscope consists of an eyepiece, an objective lens, a stage, a light source and a condenser. A compound microscope can be used to observe the structure and function of cells and tissues.
- The basic chemical constituents of living body are:
  - Water: The most abundant and essential molecule in living organisms that provides a medium for biochemical reactions, transport, temperature regulation and lubrication.
  - Carbohydrates: The main source of energy and structural components in living organisms that are composed of monosaccharides, disaccharides and polysaccharides.
  - Lipids: The diverse group of molecules that are insoluble in water and serve as energy storage, membrane components, hormones and

  signaling molecules. Lipids include fats, oils, phospholipids, steroids and waxes.
  - Proteins: The complex molecules that perform various functions in living organisms, such as catalysis, transport, movement, defense, regulation and communication. Proteins are composed of amino acids linked by peptide bonds.
  - Nucleic acids: The molecules that store and transmit genetic information in living organisms. Nucleic acids are composed of nucleotides linked by phosphodiester bonds. The two types of nucleic acids are DNA and RNA.

Hello, I am Sydney, your AI assistant. I can help you with your query. Here are some notes on the topic of the cell concept for the Unit 1 of Remedial Biology-I KCS.

The cell concept

- The cell concept is the idea that **all living things are composed of cells**, and that **cells are the basic units of life** .
- The cell concept was developed by various scientists in the 17th and 18th centuries, such as Robert Hooke, Antonie van Leeuwenhoek, Matthias Schleiden, Theodor Schwann, and Rudolf Virchow.
- The cell concept is one of the foundations of modern biology, as it helps us understand the structure, function, diversity, and evolution of living organisms .

Structure of prokaryotic and eukaryotic cells

- Prokaryotic cells are cells that **lack a nucleus and other membrane-bound organelles** . They are usually **smaller and simpler** than eukaryotic cells, and are found in bacteria and archaea .
- Eukaryotic cells are cells that **have a nucleus and other membrane-bound organelles** . They are usually **larger and more complex** than prokaryotic cells, and are found in animals, plants, fungi, and protists .
- Some of the common features of prokaryotic and eukaryotic cells are:
  - A **cell membrane** that regulates the passage of substances in and out of the cell .
  - A **cytoplasm** that contains the cell's metabolic machinery and various molecules .
  - A **genetic material** that stores and transmits the cell's information .
  - **Ribosomes** that synthesize proteins .
- Some of the differences between prokaryotic and eukaryotic cells are:
  - Prokaryotic cells have a **circular** DNA molecule, while eukaryotic cells have **linear** DNA molecules .

  - Prokaryotic cells have a **nucleoid** region where the DNA is located, while eukaryotic cells have a **nuclear envelope** that separates the DNA from the cytoplasm .
  - Prokaryotic cells have **few or no** internal membranes, while eukaryotic cells have **many** internal membranes that form various organelles .
  - Prokaryotic cells have **smaller and simpler** ribosomes, while eukaryotic cells have **larger and more complex** ribosomes .
  - Prokaryotic cells may have **flagella, pili, or capsules** for locomotion, attachment, or protection, while eukaryotic cells may have **cilia, flagella, or pseudopodia** for locomotion or feeding .

Structure of plant and animal cells

- Plant and animal cells are both examples of eukaryotic cells, but they have some **distinct features** that reflect their different functions and adaptations .
- Some of the common features of plant and animal cells are:
  - A **cell membrane** that regulates the passage of substances in and out of the cell .
  - A **nucleus** that contains the cell's genetic material and controls the cell's activities .
  - A **cytoplasm** that contains the cell's metabolic machinery and various molecules .
  - **Ribosomes** that synthesize proteins .
  - **Mitochondria** that produce energy for the cell through cellular respiration .
  - **Endoplasmic reticulum** that transports and modifies proteins and lipids .
  - **Golgi apparatus** that sorts and packages proteins and lipids for secretion or delivery .
  -

Structure of Prokaryotic Cells

- Prokaryotic cells are unicellular organisms that lack a nucleus and other membrane-bound organelles.
- Prokaryotic cells have a simple structure, consisting of the following components:
  - **Plasma membrane**: an outer covering that separates the cell's interior from its surrounding environment. It regulates the passage of substances into and out of the cell.
  - **Cytoplasm**: a gel-like substance that fills the cell and contains the genetic material and other cellular structures.

- **Nucleoid**: an irregularly-shaped region that contains the cell's DNA, which is usually circular and double-stranded. It is not surrounded by a nuclear envelope.
- **Ribosomes**: small structures that synthesize proteins using the information encoded in the DNA. They are composed of RNA and protein molecules.
- **Cell wall**: a stiff layer that maintains the cell's shape, protects the cell from mechanical damage, and prevents the cell from bursting when it takes up water. The cell wall of most bacteria contains peptidoglycan, a polymer of linked sugars and polypeptides.
- **Capsule**: an optional outer layer that covers the cell wall in some bacteria. It provides additional protection and helps the cell adhere to surfaces or other cells.
- **Flagella**: long, whip-like appendages that enable the cell to move through liquid environments. They are composed of a protein called flagellin and rotate by a motor at the base.
- **Pili**: short, hair-like projections that help the cell attach to other cells or surfaces. They can also transfer genetic material between cells through a process called conjugation.
- **Plasmids**: small, circular pieces of DNA that are separate from the nucleoid. They can carry genes that confer advantages to the cell, such as antibiotic resistance or metabolic capabilities.

Eukaryotic Cells

- Eukaryotic cells are one of the two types of cells that make up all living things. The other type is prokaryotic cells.
- Eukaryotic cells have a nucleus that contains the genetic material (DNA) and is surrounded by a nuclear membrane. Prokaryotic cells do not have a nucleus or a nuclear membrane.
- Eukaryotic cells also have other membrane-bound structures called organelles that perform specific functions. Some of the main organelles are:
  - Mitochondria: the powerhouses of the cell that produce energy (ATP) from food molecules.
  - Endoplasmic reticulum: a network of membranes that synthesize and transport proteins and lipids.
  - Golgi apparatus: a stack of flattened membranes that modify, sort, and package proteins and lipids for export or use within the cell.
  - Lysosomes: sacs of enzymes that digest and recycle materials in the cell.
  - Peroxisomes: sacs of enzymes that break down toxic substances and fatty acids in the cell.
  - Cytoskeleton: a framework of protein filaments that provide shape, support, and movement to the cell and its organelles.
  - Ribosomes: small structures that make proteins using the informa-

tion from the DNA.

- Eukaryotic cells can be divided into two major groups: plant cells and animal cells. Plant cells have some additional organelles that animal cells do not have, such as:
  - Chloroplasts: organelles that contain chlorophyll and perform photosynthesis, the process of converting light energy into chemical energy.
  - Cell wall: a rigid layer of cellulose that surrounds the cell membrane and provides support and protection to the cell.
  - Vacuole: a large sac that stores water, nutrients, and waste products in the cell.
- Eukaryotic cells communicate with each other and their environment through cell signaling, which involves the use of molecules called signals that bind to receptors on the cell membrane or inside the cell. Cell signaling can regulate various cellular processes, such as growth, division, differentiation, and death.
- Eukaryotic cells can be observed and studied using a compound microscope, which consists of two sets of lenses that magnify the image of the specimen. The compound microscope can reveal the basic structure and features of eukaryotic cells, such as the nucleus, the cytoplasm, and the cell membrane.

Plant Cells

- Plant cells are the **cells present in green plants**, photosynthetic eukaryotes of the kingdom Plantae.
- Plant cells, like animal cells, are **eukaryotic**, meaning they have a **membrane-bound nucleus** and **organelles** .
- Unlike animal cells, plant cells have a **cell wall** surrounding the cell membrane. The cell wall is a **tough layer** made of **cellulose, hemicelluloses, and pectin** that provides **structural support** and **protection** to the cell .
- Many types of plant cells contain a **large central vacuole**, a water-filled volume enclosed by a membrane known as the **tonoplast**. The vacuole serves as a **storage** for water, ions, sugars, and other substances, and also helps to **maintain turgor pressure** and **cell shape** .
- Plant cells have **chloroplasts**, specialized organelles found only in plants and some types of algae. These organelles contain **chlorophyll**, a green pigment that **absorbs light** and **converts it into chemical energy** through **photosynthesis**. Chloroplasts have their own **DNA** and **ribosomes**, and are surrounded by a **double membrane** .
- Plant cells have other organelles that are common to both plant and animal cells, such as:
  - **Nucleus**: The **control center** of the cell that contains the **genetic material** (DNA) and **regulates gene expression**.
  - **Cytoplasm**: The **fluid** that fills the cell and **suspends** the or-

ganelles.

- **Plasma membrane**: The **selective barrier** that **encloses** the cell and **controls** the **movement** of substances in and out of the cell.
- **Ribosomes**: The **sites** of **protein synthesis** that are either **free** in the cytoplasm or **attached** to the **endoplasmic reticulum** (ER).
- **Endoplasmic reticulum (ER)**: A **network** of **membranous tubules** that **transports** and **modifies** proteins and other molecules. The ER can be **smooth** (without ribosomes) or **rough** (with ribosomes).
- **Golgi apparatus**: A **stack** of **flattened membranes** that **receives**, **sorts**, and **packages** proteins and other molecules from the ER for **secretion** or **delivery** to other organelles.
- **Mitochondria**: The **powerhouses** of the cell that **produce adenosine triphosphate (ATP)**, the **universal energy currency** of the cell, through **cellular respiration**. Mitochondria have their own **DNA** and **ribosomes**, and are surrounded by a **double membrane**.
- **Lysosomes**: The **digestive** organelles that **contain hydrolytic enzymes** that **break down waste** and **foreign** materials in the cell.
- **Peroxisomes**: The **oxidative** organelles that **contain catalase** and other enzymes that **detoxify harmful** substances such as **hydrogen peroxide** and **alcohol** in the cell.
- **Cytoskeleton**: The **framework** of the cell that **provides shape**, **support**, and **movement** to the cell and its organelles. The cytoskeleton is composed of **microtubules**, **microfilaments**, and **intermediate filaments**.
- **Plasmodesmata**: The **channels** that **connect** adjacent plant cells and **allow** the **exchange** of **molecules** and **signals** between them.

Animal Cells

- Animal cells are the basic structural and functional units of animal tissues and organs .
- Animal cells are eukaryotic cells, which means they have a nucleus and membrane-bound organelles .
- Animal cells are surrounded by a plasma membrane, which forms a selective barrier that allows nutrients to enter and waste products to leave.
- Animal cells have different types and functions, such as skin cells, muscle cells, blood cells, nerve cells, and fat cells.
- Animal cells have some structures that are not found in plant cells, such as lysosomes and centrosomes.
- Animal cells do not have some structures that are found in plant cells, such as cell walls and large central vacuoles.

Structure of Animal Cells

- The structure of animal cells can be seen in the following diagram:

Animal cell diagram

- The main components of animal cells are:
  - Nucleus: The nucleus is the control center of the cell, where the genetic material (DNA) is stored and replicated. The nucleus is surrounded by a nuclear envelope, which has pores that allow the passage of molecules. The nucleus also contains a nucleolus, where ribosomes are synthesized .
  - Cytoplasm: The cytoplasm is the fluid-filled space inside the cell, where various chemical reactions take place. The cytoplasm contains various organelles and molecules that perform different functions .
  - Plasma membrane: The plasma membrane is the outer layer of the cell, which separates it from the external environment. The plasma membrane is composed of a phospholipid bilayer, with embedded proteins and carbohydrates. The plasma membrane regulates the movement of substances into and out of the cell, and also mediates cell communication and recognition .
  - Mitochondria: Mitochondria are the powerhouses of the cell, where the energy-rich molecule ATP (adenosine triphosphate) is produced by cellular respiration. Mitochondria have a double membrane, with an inner membrane folded into cristae. Mitochondria also have their own DNA and ribosomes .
  - Ribosomes: Ribosomes are the sites of protein synthesis, where the genetic information from the DNA is translated into amino acid sequences. Ribosomes can be found either free in the cytoplasm or attached to the endoplasmic reticulum .
  - Endoplasmic reticulum (ER): The endoplasmic reticulum is a network of membranous tubules and sacs that extends from the nuclear envelope. The ER has two types: rough ER and smooth ER. The rough ER has ribosomes attached to it, and is involved in protein synthesis and modification. The smooth ER does not have ribosomes, and is involved in lipid synthesis and detoxification .
  - Golgi apparatus: The Golgi apparatus is a stack of flattened membranous sacs that receives, modifies, sorts, and packages proteins and lipids from the ER. The Golgi apparatus also produces vesicles that transport materials to different destinations within or outside the cell .
  - Lysosomes: Lysosomes are vesicles that contain digestive enzymes that break down various substances, such as food particles, bacte-

ria, viruses, and worn-out organelles. Lysosomes also play a role in apoptosis (programmed cell death) .

  - Centrosomes: Centrosomes are structures that consist of two centrioles, which are cylindrical arrays of microtubules. Centrosomes are involved in organizing the cytoskeleton and the spindle fibers during cell division .

  - Cytoskeleton: The cytoskeleton is a network of protein filaments and tubules that provide shape, support, and movement to the cell. The cytoskeleton consists of three types of fibers: microfilaments, intermediate filaments

Structure and function of cell membrane

- The cell membrane, also called the plasma membrane, is found in all cells and separates the interior of the cell from the outside environment.
- The cell membrane consists of a lipid bilayer that is semipermeable, meaning it allows some substances to cross, but not others.
- The cell membrane regulates the transport of materials entering and exiting the cell by various mechanisms, such as diffusion, osmosis, facilitated diffusion, active transport, endocytosis, and exocytosis.
- The cell membrane also protects the integrity of the cell, supports the cell, and helps to maintain the cell's shape.
- The cell membrane contains various types of proteins that have different functions, such as:
  - Structural proteins that help to give the cell support and shape.
  - Transport proteins that form channels or carriers for passage of materials across the membrane .
  - Receptor proteins that help cells communicate with their external environment through the use of hormones, neurotransmitters, and other signaling molecules.
  - Enzymatic proteins that catalyze chemical reactions on the membrane surface or within the cell.
  - Marker proteins that provide identification markers for the cell and allow recognition by other cells or molecules, such as the immune system .
- The cell membrane also contains cholesterol molecules that are interspersed among the phospholipids and help to modulate the fluidity and stability of the membrane.
- The cell membrane also contains carbohydrate groups that are attached to some of the lipids and proteins and serve as recognition sites for cell-cell interactions, such as cell adhesion and cell signaling.

Cell organelles and their function

- Cell organelles are small structures within the cytoplasm that carry out functions necessary to maintain homeostasis in the cell.
- Cell organelles are involved in many processes, such as energy production, protein synthesis, secretion, digestion, detoxification, and signal transduction .
- Cell organelles can be classified into two types: membranous and non-membranous.
- Membranous organelles are surrounded by a lipid bilayer membrane, similar to the plasma membrane that encloses the cell.
- Non-membranous organelles are not enclosed by a membrane and are directly in contact with the cytoplasm.
- Some of the major cell organelles and their functions are:
  - Plasma membrane: It is the outermost layer of the cell that separates it from the external environment. It is selectively permeable, meaning it allows some substances to pass through while blocking others. It is involved in transport, communication, and recognition .
  - Cytoplasm: It is the jelly-like substance that fills the cell and contains the organelles. It provides a medium for chemical reactions and molecular movements .
  - Nucleus: It is the largest and most prominent organelle in eukaryotic cells. It contains the genetic material (DNA) of the cell and controls its activities. It is surrounded by a double membrane called the nuclear envelope, which has pores for the exchange of materials. It also contains a dense structure called the nucleolus, which is the site of ribosome synthesis .
  - Mitochondria: They are the powerhouses of the cell, where the energy from glucose is converted into ATP (adenosine triphosphate), the universal energy currency of the cell. They have a double membrane, with the inner membrane folded into cristae that increase the surface area for energy production. They also have their own DNA and ribosomes, and can replicate independently of the nucleus .
  - Ribosomes: They are the sites of protein synthesis, where the information from mRNA (messenger RNA) is translated into amino acid sequences. They are composed of two subunits, one large and one small, made of rRNA (ribosomal RNA) and proteins. They can be found either free in the cytoplasm or attached to the endoplasmic reticulum .
  - Endoplasmic reticulum (ER): It is a network of membranous tubules and sacs that extends from the nuclear envelope to the plasma membrane. It has two types: rough ER and smooth ER. Rough ER has ribosomes attached to its surface and is involved in protein synthe-

sis and modification. Smooth ER lacks ribosomes and is involved in lipid synthesis, detoxification, and calcium storage .

  - Golgi apparatus: It is a stack of flattened membranous sacs that receives, sorts, modifies, and packages proteins and lipids from the ER. It also produces lysosomes and secretory vesicles that carry the products to their destinations .
  - Lysosomes: They are spherical membranous sacs that contain hydrolytic enzymes that digest various materials, such as food particles, bacteria, viruses, worn-out organelles, and cellular debris. They are also involved in apoptosis (programmed cell death) and autophagy (self-eating) .
  - Peroxisomes: They are similar to lysosomes, but contain different enzymes that break down fatty acids, amino acids, and toxic substances, such as hydrogen peroxide. They are also involved in the synthesis of some lipids, such as cholesterol and bile salts .
  - Cytoskeleton: It is a network of protein filaments and tubules that provide shape, support, and movement to the cell and its organelles. It has three main components: microfilaments, intermediate filaments, and microtubules. Microfilaments are made of actin and are involved in muscle contraction, cell division, and cell

Basics of cell signaling

- Cell signaling is the process of cellular communication within the body driven by cells releasing and receiving hormones and other signaling molecules.
- Cell signaling enables coordination within multicellular organisms and allows cells to respond to external stimuli .
- Cell signaling involves three stages: reception, transduction, and response .
  - Reception: A cell detects a signaling molecule from the outside of the cell. The signaling molecule binds to a specific receptor protein on the cell surface or inside the cell .
  - Transduction: When the signaling molecule binds the receptor, it changes the receptor protein in some way. This triggers a series of events or a signal relay cascade inside the cell, which may involve other molecules, enzymes, or ions .
  - Response: Finally, the signal triggers a specific cellular response, such as a change in gene expression, enzyme activity, cell division, or cell death .
- Cell signaling can be classified into different forms based on the distance and type of signaling molecules involved:
  - Paracrine signaling: Cells secrete signaling molecules that affect nearby cells. For example, growth factors that stimulate cell division in neighboring cells.

  - Autocrine signaling: Cells secrete signaling molecules that bind to their own receptors. For example, some cancer cells produce growth factors that stimulate their own growth.
  - Endocrine signaling: Cells secrete signaling molecules that travel through the bloodstream and affect distant cells. For example, hormones produced by endocrine glands that regulate metabolism, growth, and reproduction.
  - Juxtacrine signaling: Cells directly contact each other through membrane proteins or gap junctions. For example, immune cells that recognize and kill infected cells.
  - Synaptic signaling: Neurons secrete signaling molecules called neurotransmitters that cross the synaptic cleft and bind to receptors on the postsynaptic cell. For example, acetylcholine that stimulates muscle contraction.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here are some notes on the structure and use of compound microscope for the unit 1 of remedial biology-I KCS.

Structure and use of compound microscope

- A compound microscope is an instrument that magnifies objects otherwise too small to be seen, producing an image in which the object appears larger.
- A compound microscope has two or more convex lenses, one objective is used at a time, it produces 2-dimensional images, and its typical magnification is between 40x and 1000x.
- A compound microscope is mainly used for studying the structural details of cell, tissue, or sections of organs.
- The parts of a compound microscope can be classified into two: non-optical parts and optical parts .
- Non-optical parts include:
  - Base: the foot of the microscope, either U or horseshoe-shaped, that supports the whole instrument.
  - Pillar: the vertical part of the base that connects it to the arm.
  - Arm: the curved part of the microscope that holds the upper parts and is used to carry the microscope.
  - Stage: the flat platform where the specimen is placed for observation, usually with clips or a mechanical stage to hold it in place .
  - Coarse adjustment knob: the large knob on the arm that is used to move the stage up and down to bring the specimen into focus .
  - Fine adjustment knob: the smaller knob on the arm that is used to make fine adjustments to the focus of the specimen .
  - Condenser: the lens system below the stage that collects and concentrates the light from the light source and directs it to the specimen

.

  - Iris diaphragm: the device below the condenser that regulates the amount of light passing through the specimen .
  - Light source: the source of illumination for the microscope, either a mirror that reflects external light or an electric lamp .
- Optical parts include:
  - Eyepiece or ocular lens: the lens at the top of the microscope that the observer looks through, usually with a magnification of 10x .
  - Body tube: the tube that connects the eyepiece to the nosepiece .
  - Nosepiece or revolving turret: the rotating part at the lower end of the body tube that holds the objective lenses .
  - Objective lenses: the lenses attached to the nosepiece that magnify the specimen, usually with 4x, 10x, 40x, and 100x magnifications .
- The total magnification of a compound microscope is the product of the magnification of the eyepiece and the objective lens .
- The working principle of a compound microscope is that the light from the light source passes through the condenser and the iris diaphragm and reaches the specimen, which reflects some of the light and absorbs the rest. The reflected light passes through the objective lens, which forms a magnified real image of the specimen. This image is further magnified by the eyepiece, which forms a virtual image that can be seen by the observer.
- A compound microscope can be configured in different ways, such as monocular, binocular, or trinocular, depending on the number of eyepieces.
- A compound microscope can also have different types of illumination, such as brightfield, darkfield, phase contrast, or fluorescence, depending on the contrast and color of the specimen.

Basic chemical constituents of living body

- The living body is composed of various chemical elements and compounds that are essential for its structure, function, and regulation.
- About 99% of the mass of the human body is made up of six elements: oxygen, carbon, hydrogen, nitrogen, calcium, and phosphorus.
- Another five elements: potassium, sulfur, sodium, chlorine, and magnesium make up about 0.85% of the remaining mass. All 11 elements are necessary for life.
- The human body also contains trace amounts of other elements, such as iron, iodine, copper, zinc, selenium, etc. that serve specific roles in various biochemical processes.
- The most abundant chemical compound in the human body is water, which accounts for about 65% of the body mass. Water is the primary solvent found in the body and is used to regulate temperature and osmotic pressure. Water also participates in many chemical reactions, such as hydrolysis and dehydration synthesis.

- The other major chemical compounds in the human body are organic molecules, which contain carbon and hydrogen as their main elements. The four main classes of organic molecules are lipids, proteins, carbohydrates, and nucleic acids.
- Lipids are fats and oils that are insoluble in water. They are mainly used for energy storage, membrane structure, and hormone synthesis. Some examples of lipids are triglycerides, phospholipids, steroids, and cholesterol.
- Proteins are polymers of amino acids that are linked by peptide bonds. They are involved in various functions, such as enzyme catalysis, structural support, transport, movement, communication, and defense. Some examples of proteins are hemoglobin, collagen, antibodies, and actin.
- Carbohydrates are sugars and starches that are soluble in water. They are mainly used for energy production, energy storage, and cell recognition. Some examples of carbohydrates are glucose, glycogen, cellulose, and chitin.
- Nucleic acids are polymers of nucleotides that are composed of a nitrogenous base, a pentose sugar, and a phosphate group. They store and transmit genetic information and direct protein synthesis. The two types of nucleic acids are DNA and RNA.

Unit 2 - Tissues in animal and plants, Morphology, anatomy and functions of different parts of plants: Root, stem, leaf, inflorescence, flower, fruit and seed.

- Tissues are groups of cells that have a similar structure and function in a multicellular organism.
- Tissues are classified into two main types: plant tissues and animal tissues.

Plant Tissues

- Plant tissues are divided into two types: meristematic tissues and permanent tissues.
- Meristematic tissues are composed of actively dividing cells that are responsible for the growth of the plant. They are found in the tips of roots and stems, and in the cambium and cork layers.
- Permanent tissues are derived from meristematic tissues and have a fixed shape and function. They are classified into three types: simple tissues, complex tissues, and special tissues.
- Simple tissues are made of one type of cell and perform a common function. They include parenchyma, collenchyma, and sclerenchyma.
- Complex tissues are made of more than one type of cell and perform a specific function. They include xylem and phloem.
- Special tissues are modified to perform a particular function in the plant. They include secretory tissues, epidermis, cork, and laticiferous tissues.

Animal Tissues

- Animal tissues are divided into four types: epithelial tissues, connective tissues, muscle tissues, and nervous tissues.
- Epithelial tissues are composed of tightly packed cells that cover the external and internal surfaces of the body. They protect the body from infection, injury, and dehydration, and also perform functions such as secretion, absorption, and sensation. They are classified based on the shape and arrangement of the cells.
- Connective tissues are composed of cells and extracellular matrix that bind and support other tissues and organs. They include blood, bone, cartilage, adipose, and fibrous tissues.
- Muscle tissues are composed of contractile cells that enable movement of the body and its parts. They include skeletal, cardiac, and smooth muscles.
- Nervous tissues are composed of neurons and neuroglia that transmit and process information in the body. They form the brain, spinal cord, and nerves.

Morphology, anatomy and functions of different parts of plants: Root, stem, leaf, inflorescence, flower, fruit and seed.

- Root: The root is the part of the plant that anchors it to the soil and absorbs water and minerals from it. The root has three regions: the root cap, the root meristem, and the root hair. The root cap protects the root tip from damage, the root meristem contains the dividing cells that produce new root tissues, and the root hair increases the surface area for absorption. The root has two types of tissues: the cortex and the stele. The cortex is the outer layer of the root that stores food and water, and the stele is the inner core of the root that contains the xylem and phloem.
- Stem: The stem is the part of the plant that supports the leaves, flowers, and fruits, and transports water and nutrients between them. The stem has three regions: the node, the internode, and the bud. The node is the point where a leaf or a branch is attached to the stem, the internode is the segment between two nodes, and the bud is the undeveloped shoot that can grow into a leaf, a flower, or a branch. The stem has two types of tissues: the epidermis and the vascular bundles. The epidermis is the outer layer of the stem that protects it from injury and water loss, and the vascular bundles are the strands of xylem and phloem that run along the stem.
- Leaf: The leaf is the part of the plant that performs photosynthesis and transpiration. The leaf has three parts: the petiole, the blade, and the veins. The petiole is the stalk that connects the leaf to the stem, the blade is the flat and broad part of the leaf that contains the

Tissues in animal and plants

- Tissues are groups of similar cells that work together to perform a specific function.
- Tissues are found in both animals and plants, but they have different types and functions.
- The four main types of animal tissues are:
  - **Epithelial tissue**: covers the body surface and lines the internal organs and cavities. It protects, absorbs, secretes, and senses.
  - **Connective tissue**: supports and binds other tissues together. It consists of cells and extracellular matrix. It includes blood, bone, cartilage, adipose, and fibrous tissues.
  - **Muscle tissue**: can contract and relax to produce movement. It consists of long, cylindrical cells called muscle fibers. It includes skeletal, cardiac, and smooth muscles.
  - **Nervous tissue**: allows rapid communication between different parts of the body. It consists of neurons and glial cells. It includes the brain, spinal cord, and nerves.
- The three main tissue systems in plants are:
  - **Epidermis**: the outermost layer of cells that covers the plant body. It protects, prevents water loss, and allows gas exchange.
  - **Ground tissue**: the bulk of the plant body that fills the space between the epidermis and the vascular tissue. It performs various functions such as photosynthesis, storage, and support. It includes parenchyma, collenchyma, and sclerenchyma cells.
  - **Vascular tissue**: the tissue that transports water and nutrients throughout the plant. It consists of xylem and phloem. Xylem conducts water and minerals from the roots to the leaves. Phloem conducts sugars and other organic substances from the leaves to the rest of the plant.

Morphology of Plants

Morphology is the study of the physical form and external structure of plants. It is useful for the identification and classification of plants based on their similarities and differences. Plant morphology includes both the vegetative and reproductive structures of plants.

Root

The root is the part of the plant that grows underground and anchors the plant to the soil. It also absorbs water and minerals from the soil and transports them to the stem. The root has three main regions: the root cap, the root tip, and the root hairs. The root cap protects the root tip from damage and secretes a slimy substance that helps the root move through the soil. The root tip contains the apical meristem, which is the region of active cell division and growth. The root hairs are extensions of the epidermal cells that increase the surface area of the root and enhance absorption.

Some types of roots are:

- Taproot: a single, thick, main root that grows vertically downward and gives rise to smaller lateral roots. Examples: carrot, radish, beetroot.
- Fibrous root: a cluster of thin, branching roots that spread out in all directions. Examples: grass, wheat, rice.
- Adventitious root: roots that arise from any part of the plant other than the radicle (the first root that emerges from the seed). Examples: prop roots of maize, aerial roots of orchids, pneumatophores of mangroves.

Stem

The stem is the part of the plant that grows above the ground and supports the leaves, flowers, and fruits. It also conducts water and minerals from the root to the leaves and food from the leaves to the rest of the plant. The stem has three main tissues: the epidermis, the vascular tissue, and the ground tissue. The epidermis is the outermost layer of cells that protects the stem and prevents water loss. The vascular tissue consists of xylem and phloem, which are the tubes that transport water and food, respectively. The ground tissue is the bulk of the stem that fills the space between the epidermis and the vascular tissue.

Some types of stems are:

- Herbaceous stem: a soft, green, flexible stem that usually dies at the end of the growing season. Examples: sunflower, tomato, mint.
- Woody stem: a hard, brown, rigid stem that persists for many years and grows in thickness. Examples: oak, pine, mango.
- Modified stem: stems that perform functions other than support and conduction. Examples: rhizome (underground stem that stores food and produces new plants), tuber (enlarged tip of a rhizome that stores food), bulb (short, vertical stem with fleshy leaves that store food), corm (short, vertical stem with dry leaves that store food), runner (horizontal stem that grows along the ground and produces new plants), stolon (horizontal stem that grows above the ground and produces new plants), tendril (coiled stem that helps the plant to climb), thorn (sharp, woody stem that protects the plant from herbivores).

Leaf

The leaf is the part of the plant that is mainly involved in photosynthesis, the process of making food from light, water, and carbon dioxide. The leaf has a flat, broad surface called the blade and a stalk called the petiole that attaches the blade to the stem. The blade has a network of veins that carry water and food to and from the leaf. The blade also has tiny pores called stomata that allow gas exchange and transpiration (loss of water vapor) to occur. The leaf has three main tissues: the epidermis, the mesophyll, and the vascular tissue. The epidermis is the outermost layer of cells that covers the upper and lower surfaces of the leaf. The mesophyll is the middle layer of cells that contains chloroplasts, the organelles that perform photosynthesis. The vascular tissue consists of xylem and phloem, which are the tubes that transport water and food, respectively.

Some types of leaves are:

- Simple leaf: a leaf that has a single blade. Examples: mango, rose, neem.
- Compound leaf: a leaf that has more than one blade, called leaflets, attached to a common petiole. Examples: pea, tamarind, fern.
- Modified leaf: leaves that perform functions other than photosynthesis. Examples: spines (reduced leaves that protect the plant from herbivores and reduce water loss), scales

Anatomy and Functions of Different Parts of Plants

Root

- The root is located below the ground and anchors the plant to the soil.
- The root absorbs water and minerals from the soil and transports them to the stem.
- The root also stores food and hormones for the plant.
- The root can be classified into two types: taproot and fibrous root.
- Taproot is the main root that grows vertically downwards and gives rise to lateral branches called secondary roots.
- Fibrous root is a network of thin and branched roots that spread horizontally in the soil.
- The root has three regions: root cap, root meristem, and root hair.
- Root cap is a protective layer of cells that covers the tip of the root and secretes mucilage to lubricate the soil.
- Root meristem is a region of actively dividing cells that produces new root cells.
- Root hair is an extension of the epidermal cell that increases the surface area for absorption.

Stem

- Stem is the part of the plant that grows above the ground and supports the leaves, flowers, and fruits.
- Stem conducts water and minerals from the root to the leaves and food from the leaves to the rest of the plant.

- Stem also produces hormones and secondary metabolites for the plant.
- Stem can be classified into two types: herbaceous stem and woody stem.
- Herbaceous stem is soft and green and usually dies at the end of the growing season.
- Woody stem is hard and brown and persists for many years.
- The stem has three tissues: epidermis, vascular tissue, and ground tissue.
- Epidermis is the outermost layer of cells that protects the stem from water loss and infection.
- Vascular tissue is the tissue that transports water and food in the stem and consists of xylem and phloem.
- Xylem is the tissue that carries water and minerals from the root to the leaves.
- Phloem is the tissue that carries food from the leaves to the rest of the plant.
- Ground tissue is the tissue that fills the space between the epidermis and the vascular tissue and provides support and storage for the stem.

Leaf

- Leaf is the part of the plant that is usually green, flat, and attached to the stem.
- Leaf is the main organ of photosynthesis, the process of converting light energy into chemical energy.
- Leaf also regulates gas exchange, transpiration, and temperature for the plant.
- Leaf can be classified into two types: simple leaf and compound leaf.
- Simple leaf is a single blade with a petiole (stalk) that connects it to the stem.
- Compound leaf is a leaf that is divided into several leaflets that are attached to a common petiole.
- The leaf has three layers: cuticle, mesophyll, and stomata.
- Cuticle is a waxy layer that covers the epidermis and prevents water loss and infection.
- Mesophyll is the middle layer of the leaf that contains chloroplasts, the organelles that perform photosynthesis.
- Stomata are the pores on the lower epidermis that allow gas exchange and transpiration.

Inflorescence

- Inflorescence is the part of the plant that bears the flowers.
- Inflorescence is the reproductive organ of flowering plants.
- Inflorescence can be classified into two types: racemose and cymose.
- Racemose inflorescence is an elongated axis with flowers arranged in an acropetal order, meaning the oldest flowers are at the base and the youngest flowers are at the tip.

- Cymose inflorescence is a shortened axis with flowers arranged in a basipetal order, meaning the oldest flowers are at the tip and the youngest flowers are at the base.
- The inflorescence has four parts: peduncle, bract, pedicel, and flower.
- Peduncle is the main stalk that supports the inflorescence.
- Bract is a modified leaf that

Root

Definition

- A root is the **descending portion of the plant axis** that is usually **underground**.
- It is one of the main organs of a vascular plant, along with the stem, leaf, flower and fruit.
- It is primarily responsible for the **fixation and absorption of water and minerals** from the soil and their **transport to the stem**.
- It also performs other functions such as **storing reserve food material**, **synthesizing plant growth regulators** and **communicating with other plants** .

Morphology

- The root with its branches is known as the **root system**.
- There are two types of root systems: **taproot system** and **fibrous root system**.
- A taproot system consists of a **single main root** that grows vertically downward and gives rise to **lateral roots** or **branches**.
- A fibrous root system consists of **many fine roots** that arise from the **base of the stem** or the **nodes of the stem** and spread in different directions.
- The root system can be **modified** for various purposes such as **storage**, **support**, **respiration**, **reproduction** and **protection**.
- Some examples of modified roots are **tuberous roots**, **prop roots**, **pneumatophores**, **suckers** and **stilt roots**.

Anatomy

- The root is composed of three main regions: the **root cap**, the **root meristem** and the **root hair zone**.
- The root cap is a **protective structure** that covers the tip of the root and secretes **mucilage** to lubricate the soil.
- The root meristem is a **region of active cell division** that produces new cells for root growth.
- The root hair zone is a **region of cell elongation and differentiation** that forms the **primary tissues** of the root.

- The primary tissues of the root are arranged in a **radial** or **concentric** pattern and include the **epidermis**, the **cortex**, the **endodermis**, the **pericycle**, the **vascular tissue** and the **pith**.
- The epidermis is the **outermost layer** of the root that forms **root hairs** for absorption of water and minerals.
- The cortex is the **thick layer** of the root that consists of **parenchyma cells** that store **starch** and other substances.
- The endodermis is the **innermost layer** of the cortex that forms a **selective barrier** between the cortex and the vascular tissue.
- The pericycle is the **thin layer** of the root that lies between the endodermis and the vascular tissue and gives rise to **lateral roots** and **secondary growth**.
- The vascular tissue is the **central core** of the root that consists of **xylem** and **phloem** for transport of water and organic materials.
- The pith is the **innermost part** of the vascular tissue that consists of **parenchyma cells** that may or may not be present in the root.

Function

- The root performs four main functions: **absorption**, **anchorage**, **storage** and **synthesis**.
- Absorption: The root absorbs water and minerals from the soil through the **root hairs** and the **epidermis** and transports them to the stem through the **xylem**.
- Anchorage: The root provides a proper **anchorage** to the plant parts by penetrating deep into the soil and forming a **firm base** for the stem.
- Storage: The root stores reserve food material such as **starch**, **sugars**, **proteins** and **fats** in the **cortex** and the **pith** cells and supplies them to the plant when needed.
- Synthesis: The root synthesizes plant growth regulators such as **auxins**, **cytokinins**, **gibberellins** and **ethylene** that regulate various aspects of plant growth and development[[1]

Stem

- The stem is an **axial organ** of the shoot system that supports and elevates the leaves, fruits, and flowers.
- The stem arranges leaves in a way that they get direct sunlight to perform photosynthesis.
- The stem also conducts water and minerals from the roots to the leaves and other parts of the plant through the **xylem** .
- The stem also transports organic substances such as sugars and hormones from the leaves to the roots and other parts of the plant through the **phloem** .
- The stem can also store food and water in some plants, such as cacti and

potatoes.

- The stem can also perform photosynthesis in some plants, such as succulents and cacti, that have green stems.

Morphology of the Stem

- The stem bears **nodes** and **internodes**.
- The nodes are the regions of the stem where leaves are born.
- The internodes are the portions between two nodes.
- The stem also bears **buds**, which are undeveloped or embryonic shoots that can give rise to branches, leaves, or flowers.
- The buds can be **terminal** or **axillary**.
- The terminal buds are located at the tip of the stem and are responsible for the elongation of the stem.
- The axillary buds are located in the axils of the leaves and can give rise to lateral branches.
- The stem can be classified into different types based on the shape, size, and mode of branching.
- Some of the common types of stems are:
  - **Herbaceous stem**: It is soft, green, and flexible, and usually dies at the end of the growing season. Examples are grasses, wheat, and sunflower.
  - **Woody stem**: It is hard, brown, and rigid, and usually persists for many years. Examples are trees and shrubs.
  - **Climbing stem**: It is weak and slender, and needs support to grow vertically. Examples are vines, creepers, and lianas.
  - **Creeping stem**: It is horizontal and grows along the surface of the soil. Examples are strawberry, mint, and grass.
  - **Tuberous stem**: It is swollen and modified for storage of food. Examples are potato, ginger, and turmeric.
  - **Bulbous stem**: It is short and reduced, and surrounded by fleshy scales. Examples are onion, garlic, and tulip.
  - **Corm stem**: It is solid and enlarged, and covered by dry scales. Examples are gladiolus, crocus, and colocasia.
  - **Rhizome stem**: It is underground and branched, and bears nodes and internodes. Examples are ginger, turmeric, and banana.

Anatomy of the Stem

- The stem is composed of three types of tissues: **dermal tissue**, **vascular tissue**, and **ground tissue** .

- The dermal tissue covers and protects the stem .
- The dermal tissue consists of the **epidermis**, which is a single layer of cells that secretes a waxy **cuticle** to prevent water loss .
- The epidermis also bears **stomata**, which are pores for gas exchange and transpiration .
- The epidermis also bears **trichomes**, which are hair-like structures that can provide insulation, defense, or secretion .
- The vascular tissue transports water and organic substances throughout the plant .
- The vascular tissue consists of the **xylem** and the **phloem**, which are arranged in bundles .
- The xylem conducts water and minerals from the roots to the leaves and other parts of the plant [[6]

Leaf

- A leaf is a thin, flat organ of the plant that is mainly responsible for photosynthesis and transpiration .
- A leaf develops laterally at the node of the stem and originates from the shoot apical meristem .
- The main function of a leaf is to produce food for the plant by photosynthesis. Chlorophyll, the substance that gives plants their characteristic green colour, absorbs light energy .
- The internal structure of the leaf is protected by the leaf epidermis, which is continuous with the stem epidermis. The leaf epidermis may have stomata, which are pores that allow gas exchange and water loss .
- The leaf mesophyll is the tissue between the upper and lower epidermis that contains the photosynthetic cells. The mesophyll may be divided into palisade and spongy layers, depending on the arrangement and shape of the cells .
- The leaf veins are the vascular bundles that transport water and nutrients to and from the leaf. The veins may form different patterns, such as parallel, netted, or dichotomous, depending on the type of plant .
- The leaf margin is the edge of the leaf blade. The margin may have different shapes, such as entire, serrated, lobed, or crenate, depending on the type of plant .
- The leaf base is the part of the leaf that attaches to the stem. The leaf base may have different structures, such as stipules, sheath, or petiole, depending on the type of plant .
- The leaf apex is the tip of the leaf blade. The apex may have different shapes, such as acute, obtuse, acuminate, or mucronate, depending on the type of plant .
- Leaves may be modified for different purposes, such as storage, protection, support, or reproduction. Some examples of leaf modifications are spines, tendrils, bracts, and scales .

- Leaves may be classified into different types, based on the number, shape, and arrangement of the leaflets. Some examples of leaf types are simple, compound, pinnate, palmate, and whorled .

Inflorescence

- An inflorescence is the arrangement of a cluster of flowers on a floral axis or peduncle.
- Inflorescence can be classified into different types based on the arrangement of flowers and the timing of flowering .
- The main types of inflorescence are:
  - Racemose inflorescence (Indeterminate inflorescence): The flowers are arranged along the peduncle in an acropetal order, i.e., the older flowers are at the base and the younger ones are at the apex. The peduncle continues to grow and produce flowers .
  - Cymose inflorescence (Determinate inflorescence): The flowers are arranged along the peduncle in a basipetal order, i.e., the older flowers are at the apex and the younger ones are at the base. The peduncle terminates in a flower and stops growing .
  - Special types of inflorescence: These are modified forms of racemose or cymose inflorescence that have a characteristic shape or structure. Some examples are capitulum, spadix, catkin, cyathium, etc .
  - Mixed inflorescence: These are combinations of two or more types of inflorescence on the same plant. Some examples are verticillaster, thyrsus, anthodium, etc.
- Inflorescence is important for the reproduction and dispersal of plants. It also influences the pollination and fruiting of plants. Inflorescence can also be used for identification and classification of plants .

Flower

- A flower is a reproductive structure of a flowering plant (angiosperm) that produces seeds.
- A flower consists of a floral axis that bears the essential organs of reproduction (stamens and pistils) and usually accessory organs (sepals and petals).
- The stamens are the male parts of the flower that produce pollen, which contains the sperm cells. The pistils are the female parts of the flower that produce ovules, which contain the egg cells.
- The sepals are the outermost whorl of the flower that usually protect the flower bud before it opens. The petals are the next whorl of the flower that usually attract pollinators with their color and scent.
- The floral axis may have a receptacle, which is the enlarged end of the stem that supports the flower parts. The floral axis may also have a pedicel,

which is the stalk that connects the flower to the stem.

- The arrangement of the flower parts on the floral axis is called the floral symmetry. There are two main types of floral symmetry: radial and bilateral. Radial symmetry means that the flower can be divided into equal halves along any plane passing through the center. Bilateral symmetry means that the flower can be divided into equal halves along only one plane.
- The number and arrangement of the sepals, petals, stamens, and pistils on the floral axis are called the floral formula. The floral formula can be used to identify and classify the flower.
- The function of the flower is to facilitate sexual reproduction by allowing the transfer of pollen from the stamens to the pistils, which is called pollination. Pollination can be done by wind, water, animals, or self-pollination. After pollination, the pollen grains germinate on the stigma and grow a pollen tube that delivers the sperm cells to the ovules, which is called fertilization. After fertilization, the ovules develop into seeds and the ovary develops into a fruit. The fruit protects and disperses the seeds, which can grow into new plants.

Fruit

- A fruit is the **soft, pulpy part of a flowering plant that contains seeds** .
- It is formed from the **ovaries of angiosperms** (plants that produce flowers and seeds enclosed in a carpel) and is exclusive only to this group of plants .
- Fruits are a characteristic of flowering plants. As soon as **pollen and fertilization occur**, the ovary of a plant becomes a fruit and the ovules become a seed.
- Fruits are classified into four types namely: **simple, aggregate, multiple, and accessory** .
  - Simple fruits are derived from a single ovary of a single flower, such as apples, oranges, and cherries .
  - Aggregate fruits are formed from several ovaries of a single flower, such as raspberries, blackberries, and strawberries .
  - Multiple fruits are developed from the ovaries of several flowers that fuse together, such as pineapples, figs, and mulberries .
  - Accessory fruits are fruits that include other parts of the flower besides the ovary, such as pears, peaches, and watermelons .
- Fruits have various functions in the life cycle of plants, such as **protecting the seeds, dispersing the seeds, attracting pollinators, and providing food** for animals and humans .
  - Protecting the seeds: Fruits enclose the seeds and prevent them from drying out, being eaten, or being infected by pathogens.
  - Dispersing the seeds: Fruits aid in the movement of seeds to new

  locations, either by wind, water, or animals.
  - Attracting pollinators: Fruits often have bright colors, sweet smells, or nectar to lure insects, birds, or bats that can transfer pollen to other flowers.
  - Providing food: Fruits are a source of nutrients, vitamins, and antioxidants for animals and humans, and also play a role in cultural and culinary traditions .

Unit 2 - Tissues in animal and plants, Morphology, anatomy and functions of different parts of plants: Root, stem, leaf, inflorescence, flower, fruit and seed.

Tissues in animal and plants

- A tissue is a group of similar cells that work together to perform a specific function.
- Tissues are the building blocks of organs and organ systems in multicellular organisms.
- There are two main types of tissues: plant tissues and animal tissues.

Plant tissues

- Plant tissues are classified into two types: meristematic and permanent.
- Meristematic tissues are composed of actively dividing cells that are responsible for the growth of the plant. They are found in the tips of roots and shoots, and in the cambium layer of stems and roots.
- Permanent tissues are composed of mature cells that have stopped dividing and have a specific function. They are further classified into three types: simple, complex and special.
- Simple tissues are made of one type of cell and perform a common function. They include parenchyma, collenchyma and sclerenchyma.
- Complex tissues are made of more than one type of cell and perform a specific function. They include xylem and phloem, which are involved in the transport of water and nutrients in the plant.
- Special tissues are modified to perform a particular function. They include epidermis, cork, secretory tissues and reproductive tissues.

Animal tissues

- Animal tissues are classified into four types: epithelial, connective, muscular and nervous.
- Epithelial tissues are composed of tightly packed cells that cover the surfaces of the body and line the cavities and ducts. They provide protection,

secretion, absorption and sensation. They are classified based on the shape and arrangement of the cells.
- Connective tissues are composed of cells and extracellular matrix that bind and support other tissues and organs. They include blood, bone, cartilage, adipose, tendons and ligaments.
- Muscular tissues are composed of cells that can contract and relax to produce movement. They are classified into three types: skeletal, cardiac and smooth.
- Nervous tissues are composed of neurons and neuroglia that transmit and process information. They form the brain, spinal cord and nerves.

Morphology, anatomy and functions of different parts of plants: Root, stem, leaf, inflorescence, flower, fruit and seed.

- Morphology is the study of the external form and structure of plants.
- Anatomy is the study of the internal structure and organization of plants.
- Function is the role or purpose of a plant part in the life cycle and survival of the plant.

Root

- The root is the part of the plant that anchors it to the soil and absorbs water and minerals from the soil.
- The root can be classified into two types: taproot and fibrous root.
- Taproot is the main root that grows vertically downwards from the stem and gives rise to lateral branches called secondary roots. It is found in dicot plants, such as carrot and radish.
- Fibrous root is a network of thin and branched roots that spread horizontally in the soil. It is found in monocot plants, such as grass and wheat.
- The root has three regions: root cap, root meristem and root hair zone.
- Root cap is the protective layer of cells that covers the tip of the root and secretes mucilage to lubricate the soil.
- Root meristem is the region of actively dividing cells that produces new root cells and elongates the root.
- Root hair zone is the region of root cells that have thin outgrowths called root hairs, which increase the surface area for absorption of water and minerals.
- The root has two types of tissues: primary and secondary.
- Primary tissues are derived from the apical meristem and form the primary structure of the root. They include epidermis, cortex, endodermis, pericycle, vascular cylinder and pith.
- Epidermis is the outermost layer of cells that protects the root and absorbs water and minerals through the root hairs.
- Cortex is the layer of parenchyma cells that stores food and water and provides support to the root.

- Endodermis is the innermost layer of the cortex that regulates the passage of water and minerals into the vascular cylinder. It has a band of suberin called casparian strip that blocks the apoplastic pathway of water movement.
- Pericycle is the layer of meristematic cells that surrounds the vascular cylinder and gives rise to lateral roots and secondary tissues.
- Vascular cylinder is the central

Unit 3 - Classification of living organisms

- Classification of living organisms is the scientific process of arranging organisms into different groups and subgroups based on their similarities and differences .
- Classification helps to study the diversity of life forms in a systematic and organized manner.
- Classification also helps to understand the evolutionary relationships among different groups of organisms.

Five kingdom classification

- The five kingdom classification is a system of classifying living organisms into five major groups based on their characteristics.
- The five kingdoms are:
  - **Monera**: This kingdom includes all the prokaryotic organisms, such as bacteria and cyanobacteria (blue-green algae). They are unicellular, lack a true nucleus and membrane-bound organelles, and have a simple cell structure.
  - **Protista**: This kingdom includes all the eukaryotic organisms that are not animals, plants, or fungi. They are mostly unicellular, but some are colonial or multicellular. They have a true nucleus and membrane-bound organelles, and have a diverse mode of nutrition and locomotion.
  - **Fungi**: This kingdom includes all the eukaryotic organisms that are heterotrophic and feed on organic matter by absorption. They are mostly multicellular, except for yeasts, which are unicellular. They have a true nucleus and membrane-bound organelles, and have a cell wall made of chitin.
  - **Plantae**: This kingdom includes all the eukaryotic organisms that are autotrophic and produce their own food by photosynthesis. They are multicellular, have a true nucleus and membrane-bound organelles, and have a cell wall made of cellulose.
  - **Animalia**: This kingdom includes all the eukaryotic organisms that are heterotrophic and feed on other organisms by ingestion. They are multicellular, have a true nucleus and membrane-bound organelles,

and lack a cell wall.

Major groups and principles of classification in each kingdom

- The major groups and principles of classification in each kingdom are based on the following criteria:
  - **Cell structure**: The presence or absence of a true nucleus and membrane-bound organelles, the type of cell wall, and the number of cells are used to classify organisms into prokaryotes or eukaryotes, and into unicellular or multicellular.
  - **Mode of nutrition**: The source and mode of obtaining food are used to classify organisms into autotrophs or heterotrophs, and into producers, consumers, or decomposers.
  - **Mode of locomotion**: The presence or absence of specialized structures for movement, such as flagella, cilia, pseudopodia, or limbs, are used to classify organisms into motile or non-motile.
  - **Reproduction**: The mode and type of reproduction, such as asexual or sexual, and the presence or absence of specialized organs or structures for reproduction, such as spores, seeds, or flowers, are used to classify organisms into different groups.
  - **Phylogeny**: The evolutionary history and relationships among different groups of organisms are used to classify organisms into different levels of hierarchy, such as domain, kingdom, phylum, class, order, family, genus, and species .

Systematic and binomial system of nomenclature

- Systematic is the branch of biology that deals with the identification, naming, and classification of living organisms.
- Binomial system of nomenclature is the universal system of naming living organisms using two Latin words, the first one indicating the genus and the second one indicating the species.
- The binomial system of nomenclature was developed by Carl Linnaeus, a Swedish botanist, in the 18th century.
- The rules of binomial nomenclature are:
  - The name of the genus should be capitalized and the name of the species should be lowercase.
  - The name should be written in italics or underlined when handwritten.
  - The name should be followed by the name of the author who first described the organism, or an abbreviation of it.
  - The name should be unique and universal for each organism.

- For example, the scientific name of human is *Homo sapiens* L., where *Homo* is the genus

Classification of living organisms

- Classification of living organisms is the scientific process of arranging living things into different groups and subgroups based on their similarities and differences .
- Classification helps to study the diversity of life forms in an organized and systematic way .
- Classification also helps to understand the evolutionary relationships among different groups of organisms .
- The most widely used system of classification is the Linnaean system, developed by Carl Linnaeus in the 18th century .
- The Linnaean system uses a hierarchical structure of seven major levels: kingdom, phylum, class, order, family, genus and species .
- Each level is more specific and more closely related than the previous one. For example, all animals belong to the kingdom Animalia, but only some animals belong to the phylum Chordata, and only some chordates belong to the class Mammalia, and so on .
- The lowest level of classification is the species, which is a group of organisms that can interbreed and produce fertile offspring .
- The scientific name of a species is given by the binomial system of nomenclature, which consists of two words: the genus name and the species name .
- The genus name is capitalized and the species name is lowercase. Both names are italicized or underlined. For example, the scientific name of human is Homo sapiens .
- The five kingdom classification is a system of dividing living organisms into five major groups: Animalia, Plantae, Fungi, Protista and Monera.
- The kingdom Animalia includes all multicellular animals that are heterotrophic (obtain food from other organisms) and have a nervous system and a body cavity.
- The kingdom Plantae includes all multicellular plants that are autotrophic (make their own food by photosynthesis) and have a cell wall and chloroplasts.
- The kingdom Fungi includes all multicellular or unicellular organisms that are heterotrophic and decompose organic matter. They have a cell wall made of chitin and lack chloroplasts.
- The kingdom Protista includes all unicellular or colonial organisms that are eukaryotic (have a nucleus and other membrane-bound organelles). They can be autotrophic or heterotrophic and have diverse modes of locomotion and reproduction.
- The kingdom Monera includes all unicellular organisms that are prokaryotic (lack a nucleus and other membrane-bound organelles). They have a

cell wall made of peptidoglycan and can be autotrophic or heterotrophic. They include bacteria and cyanobacteria (blue-green algae).

- The concept of animal and plant classification is based on the morphological, anatomical, physiological and biochemical characteristics of the organisms in each kingdom.
- Animal classification is further divided into several phyla, such as Chordata, Arthropoda, Mollusca, Annelida, Nematoda, Cnidaria, Porifera, etc.
- Plant classification is further divided into several divisions, such as Bryophyta, Pteridophyta, Gymnospermae, Angiospermae, etc.
- Each phylum or division is further subdivided into classes, orders, families, genera and species, following the Linnaean system.

Five Kingdom Classification

- Five kingdom classification is a system of grouping living organisms into five major categories based on their characteristics such as cell structure, mode of nutrition, reproduction, phylogenetic relationships, and thallus organization .
- The five kingdoms are: Monera, Protista, Fungi, Plantae, and Animalia.
- This classification was proposed by Robert H. Whittaker in 1969 and is widely used in textbooks and scientific literature .

Kingdom Monera

- Kingdom Monera includes all the prokaryotic organisms, which are unicellular and lack a true nucleus and membrane-bound organelles.
- Monerans can be autotrophic or heterotrophic, aerobic or anaerobic, and can live in various habitats such as soil, water, and human body.
- Monerans reproduce mainly by binary fission, but some can also exchange genetic material by conjugation, transformation, or transduction.
- Monerans are classified into two major groups: Archaebacteria and Eubacteria.
- Archaebacteria are ancient bacteria that can survive in extreme environments such as hot springs, salt lakes, and deep-sea vents. They have unique cell wall and membrane composition and metabolic pathways.
- Eubacteria are the most common and diverse bacteria that can be found in almost every environment. They have a peptidoglycan cell wall and a variety of shapes and arrangements. They can be further classified into subgroups based on their staining properties, such as Gram-positive, Gram-negative, and acid-fast bacteria.

Kingdom Protista

- Kingdom Protista includes all the eukaryotic organisms that are unicellular or colonial and do not fit into the other kingdoms.

- Protists can be autotrophic or heterotrophic, motile or non-motile, and can have various modes of reproduction such as binary fission, multiple fission, budding, spore formation, and sexual reproduction.
- Protists are classified into several groups based on their characteristics, such as:
  - Chrysophytes: They are golden algae and diatoms that have a cell wall made of silica and are mostly photosynthetic and aquatic.
  - Dinoflagellates: They are mostly marine and photosynthetic protists that have two flagella and a cellulose cell wall. Some of them can produce bioluminescence and toxins.
  - Euglenoids: They are mostly freshwater and photosynthetic protists that have one or two flagella and no cell wall. They can also switch to heterotrophic mode in the absence of light.
  - Slime molds: They are heterotrophic protists that feed on decaying organic matter and can form multicellular aggregates or fruiting bodies under certain conditions.
  - Protozoans: They are heterotrophic protists that are animal-like in their mode of nutrition and locomotion. They can be classified into four subgroups based on their locomotory organelles, such as:
    - Amoeboids: They move and feed by pseudopodia, which are extensions of their cytoplasm. They are mostly found in freshwater and marine habitats and some are parasitic.
    - Flagellates: They move and feed by one or more flagella, which are whip-like structures. They can be free-living or symbiotic and some are pathogenic.
    - Ciliates: They move and feed by cilia, which are hair-like structures. They are mostly found in freshwater and marine habitats and have a complex body organization and life cycle.
    - Sporozoans: They are non-motile and parasitic protists that have a complex life cycle involving multiple hosts and spore formation. They cause diseases such as malaria and toxoplasmosis.

Kingdom Fungi

- Kingdom Fungi includes all the eukaryotic organisms that are multicellular (except yeasts) and heterotrophic by absorption, which means they secrete enzymes to digest their food externally and then absorb the nutrients.
- Fungi have a cell wall made of chitin and a body composed of filaments called hyphae, which can form a network called mycelium.
- Fungi reproduce by spores, which can be produced asexually or sexually. Some fungi can also reproduce by fragmentation or budding.
- Fungi are classified into four major groups based on their

Major Groups for the Notes of the Unit 3 - Classification of Living Organisms

Five Kingdom Classification

- The five kingdom classification was proposed by **Robert.H. Whittaker** in 1969 .
- This classification was based on certain characters like **mode of nutrition, thallus organization, cell structure, phylogenetic relationships and reproduction** .
- This form of kingdom classification includes **five kingdoms**: Monera, Protista, Fungi, Plantae and Animalia .

Kingdom Monera

- This kingdom includes all the **prokaryotic organisms**, such as bacteria, cyanobacteria, archaebacteria and mycoplasma .
- They have a **simple cell structure** with no membrane-bound organelles or nucleus .
- They can be **autotrophic** (photosynthetic or chemosynthetic) or **heterotrophic** (saprophytic or parasitic) in their mode of nutrition .
- They reproduce mainly by **asexual methods**, such as binary fission, budding, fragmentation or spore formation .
- They are found in **various habitats**, such as soil, water, air, plants, animals and extreme environments .

Kingdom Protista

- This kingdom includes all the **unicellular eukaryotic organisms**, such as protozoa, algae and slime molds .
- They have a **complex cell structure** with membrane-bound organelles and nucleus .
- They can be **autotrophic** (photosynthetic) or **heterotrophic** (holozoic, saprophytic or parasitic) in their mode of nutrition .
- They reproduce by **asexual methods**, such as binary fission, multiple fission or budding, or by **sexual methods**, such as conjugation, syngamy or gametic fusion .
- They are mostly **aquatic**, either freshwater or marine, or live in moist soil or as symbionts in other organisms .

Kingdom Fungi

- This kingdom includes all the **multicellular or unicellular eukaryotic organisms** that are heterotrophic and feed on dead or decaying organic matter, such as molds, yeasts, mushrooms and lichens .
- They have a **cell wall** made of chitin and a **body** composed of hyphae that form a network called mycelium .

- They are **heterotrophic** and absorb nutrients from the substratum by secreting digestive enzymes .
- They reproduce by **asexual methods**, such as fragmentation, budding, sporulation or gemmule formation, or by **sexual methods**, such as plasmogamy, karyogamy or meiosis .
- They are found in **terrestrial habitats**, such as soil, wood, dung, food, or as parasites or symbionts in

Principles of Classification in Each Kingdom

The five kingdom classification system was proposed by R.H. Whittaker in 1969 based on the following criteria :

- Cellular structure: prokaryotic or eukaryotic
- Body organization: unicellular or multicellular
- Mode of nutrition: autotrophic or heterotrophic
- Phylogenetic relationships: evolutionary history and molecular evidence
- Reproduction: sexual or asexual

The five kingdoms are:

- Monera: prokaryotic, unicellular, autotrophic or heterotrophic, asexual reproduction, includes bacteria and cyanobacteria
- Protista: eukaryotic, mostly unicellular, autotrophic or heterotrophic, sexual or asexual reproduction, includes protozoa, algae, slime molds and water molds
- Fungi: eukaryotic, mostly multicellular, heterotrophic by absorption, sexual or asexual reproduction, includes molds, yeasts, mushrooms and lichens
- Plantae: eukaryotic, multicellular, autotrophic by photosynthesis, sexual or asexual reproduction, includes mosses, ferns, gymnosperms and angiosperms
- Animalia: eukaryotic, multicellular, heterotrophic by ingestion, sexual reproduction, includes sponges, cnidarians, worms, mollusks, arthropods, echinoderms and chordates

Each kingdom is further divided into smaller groups based on more specific characteristics, such as body symmetry, tissue organization, segmentation, appendages, etc. The hierarchy of classification is:

- Kingdom
- Phylum (or Division for plants)
- Class
- Order
- Family
- Genus
- Species

The scientific name of an organism is composed of its genus and species, and is written in italics with the first letter of the genus capitalized. For example, the scientific name of human is Homo sapiens. This system of naming is called binomial nomenclature and was developed by Carl Linnaeus in the 18th century.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some information on the systematic and binomial system of nomenclature.

Systematic and Binomial System of Nomenclature

Systematic

- Systematic is the branch of biology that deals with the identification, naming, and classification of living organisms.
- Systematic aims to organize the diversity of life into a coherent and logical framework based on their evolutionary relationships and natural affinities.
- Systematic uses various methods and sources of data, such as morphology, anatomy, embryology, genetics, biochemistry, molecular biology, and biogeography, to infer the evolutionary history and phylogeny of organisms.
- Systematic also provides a common language and a standard reference for communicating and studying the diversity of life.

Binomial Nomenclature

- Binomial nomenclature is the system of scientifically naming organisms developed by Carl Linnaeus in the 18th century.
- Binomial nomenclature is required in biology to unify the naming system throughout life sciences and, as a result, offer a single unique name identifier for a species across languages.
- Binomial nomenclature is a systematic procedure for designating species. The nomenclature is made up of two names: the genus name and the specific epithet.
- The genus name is the first word of the binomial name and is always capitalized. The specific epithet is the second word of the binomial name and is always lowercase. Both words are written in italics or underlined.
- For example, the scientific name of the human species is Homo sapiens, where Homo is the genus name and sapiens is the specific epithet.
- The binomial name is also followed by the name of the authority who first described the species and the year of publication, usually in parentheses. For example, Homo sapiens Linnaeus, 1758.
- The binomial name is unique and universal for each species and follows the rules and principles of the International Code of Zoological Nomenclature (ICZN) for animals and the International Code of Nomenclature for algae, fungi, and plants (ICN) for plants .

- The binomial name reflects the classification of the species, as the genus name indicates the group of closely related species to which the species belongs. However, the binomial name does not indicate the rank or level of the taxon in the hierarchy of classification.

Concept of animal and plant classification

- Biological classification is the scientific process of grouping living organisms into categories based on their similarities and differences.
- Classification helps to organize the diversity of life and to understand the evolutionary relationships among different groups of organisms.
- The most widely used system of classification is the Linnaean taxonomy, which was developed by the Swedish naturalist Carl Linnaeus in the 18th century.
- Linnaeus classified all living organisms into two kingdoms – Plantae (plants) and Animalia (animals) – based on their morphology and anatomy.
- He also introduced the binomial system of nomenclature, which assigns a two-part scientific name to each species, consisting of the genus name and the species name.
- For example, the scientific name of human is Homo sapiens, where Homo is the genus and sapiens is the species.
- Linnaeus also divided each kingdom into smaller groups called classes, orders, families, genera and species, based on more specific characteristics.
- For example, the class Mammalia (mammals) is divided into orders such as Primates, Carnivora, Rodentia, etc., and each order is further divided into families, genera and species.
- The Linnaean system of classification is hierarchical, meaning that each group is nested within a larger group, and each group includes all the groups below it.
- The Linnaean system of classification is also based on the principle of homology, meaning that organisms that share similar structures are more closely related than those that do not.
- For example, the forelimbs of humans, bats, whales and horses are homologous, meaning that they have the same basic structure and origin, but different functions and adaptations.
- However, the Linnaean system of classification has some limitations, such as:
  - It does not account for the diversity of microorganisms, such as bacteria, archaea, protists and fungi, which are very different from plants and animals in their structure, function and evolution.
  - It does not reflect the evolutionary history and genetic relationships of organisms, which are better revealed by molecular and biochemical evidence.
  - It does not accommodate the cases of convergent evolution, where

unrelated organisms evolve similar features due to similar environmental pressures.
  - For example, the wings of birds, bats and insects are analogous, meaning that they have different structures and origins, but similar functions and adaptations.
- Therefore, in the 20th century, several alternative systems of classification were proposed, such as the phylogenetic system, the cladistic system and the phenetic system, which use different criteria and methods to group organisms.
- One of the most influential systems of classification was the five kingdom system, which was proposed by the American biologist Robert Whittaker in 1969.
- Whittaker classified all living organisms into five kingdoms – Monera, Protista, Fungi, Plantae and Animalia – based on their cell structure, mode of nutrition and level of organization.
- The five kingdoms are briefly described as follows:
  - Monera: This kingdom includes all the prokaryotic organisms, such as bacteria and archaea, which lack a true nucleus and membrane-bound organelles. They are unicellular and have diverse modes of nutrition, such as photosynthesis, chemosynthesis, parasitism, saprotrophism, etc. They reproduce mainly by binary fission or budding.
  - Protista: This kingdom includes all the eukaryotic organisms that are not plants, animals or fungi, such as protozoa, algae, slime molds, etc. They have a true nucleus and membrane-bound organelles. They are mostly unicellular, but some are colonial or multicellular. They have diverse modes of nutrition, such as photosynthesis, heterotrophy, mixotrophy, etc. They reproduce mainly by asexual methods, such as binary fission, multiple fission, spore formation, etc., but some also reproduce sexually by conjugation, syngamy, etc.
  - Fungi: This kingdom includes all the eukaryotic organisms that are heterotrophic and absorb their food from the environment, such as mushrooms, molds, yeasts, etc. They have a true nucleus and membrane-bound organelles. They are mostly multicellular, but some are unicellular. They have a cell wall made of chitin. They reproduce mainly by asexual methods, such as fragmentation, budding, spore formation, etc., but some also reproduce sexually by plasmogamy, karyogamy, meiosis, etc.
  - Plantae: This kingdom includes all the eukary

Unit 4 - Concepts of alleles and genes, Mendelian Experiments, Cell cycle (Elementary Idea), mitosis and meiosis, techniques to study mitosis and meiosis. Origin of life, evidences for biological evolution, Mechanism of evolution —Variation (Mutation and Recombination), Natural Selection with examples, Types of natural selection

- **Alleles** are different versions of the same gene that have slightly different DNA sequences and may code for different traits. For example, the gene for eye color has different alleles that determine whether the eyes are blue, brown, green, etc.
- **Genes** are segments of DNA that code for a specific protein or trait. They are located on chromosomes, which are long strands of DNA that carry many genes. Each chromosome has a pair of homologous chromosomes, one inherited from each parent. The same gene is located at the same position (locus) on each homologous chromosome.
- **Mendelian Experiments** are a series of experiments conducted by Gregor Mendel, a monk and plant breeder, in the 19th century. He studied the inheritance of traits in pea plants and discovered the basic principles of genetics, such as dominance, recessiveness, segregation, and independent assortment.
- **Cell cycle** is the process of cell growth and division that occurs in all living cells. It consists of four phases: G1 (gap 1), S (synthesis), G2 (gap 2), and M (mitosis or meiosis). In G1, the cell grows and prepares for DNA replication. In S, the cell replicates its DNA. In G2, the cell grows and prepares for cell division. In M, the cell divides into two daughter cells, either by mitosis (for somatic cells) or by meiosis (for gametes).
- **Mitosis** is a type of cell division that produces two genetically identical daughter cells from one parent cell. It is used for growth, repair, and asexual reproduction. It consists of four stages: prophase, metaphase, anaphase, and telophase. In prophase, the chromosomes condense and the nuclear envelope breaks down. In metaphase, the chromosomes align at the equator of the cell. In anaphase, the sister chromatids separate and move to opposite poles of the cell. In telophase, the chromosomes decondense and the nuclear envelope reforms. Cytokinesis, the division of the cytoplasm, usually occurs simultaneously with telophase.
- **Meiosis** is a type of cell division that produces four genetically different daughter cells from one parent cell. It is used for sexual reproduction and reduces the chromosome number by half. It consists of two rounds of division: meiosis I and meiosis II. In meiosis I, the homologous chromosomes pair up and exchange segments (crossing over) in prophase I, then separate in anaphase I. This results in two haploid cells, each with one chromosome

from each homologous pair. In meiosis II, the sister chromatids separate in anaphase II. This results in four haploid cells, each with one chromatid from each chromosome.

- **Techniques to study mitosis and meiosis** include microscopy, staining, and karyotyping. Microscopy is the use of a microscope to observe the cells and their structures. Staining is the use of dyes or chemicals to enhance the contrast or visibility of the chromosomes or other cell components. Karyotyping is the arrangement of the chromosomes according to their size, shape, and number. It can reveal abnormalities or variations in the chromosome number or structure.
- **Origin of life** is the scientific question of how life emerged from non-living matter on Earth. There are various hypotheses and experiments that attempt to explain or simulate the origin of life, such as the primordial soup, the RNA world, the iron-sulfur world, the hydrothermal vents, etc. However, there is no definitive answer or evidence for the origin of life yet.
- **Evidences for biological evolution** are the observations and data that support the theory of evolution, which states that all living organisms share a common ancestor and have diversified over time through natural selection and other mechanisms. Some of the evidences for biological evolution are:
  - Fossil record: The preserved remains or impressions of organisms that lived in the past. They show the changes and transitions in the morphology, diversity, and distribution of life over time.
  - Comparative anatomy: The study of the similarities and differences in the body structures of different organisms. It reveals the homologies (shared traits due to common ancestry) and analogies (similar traits due to convergent evolution) among organisms.
  - Comparative embryology

Concepts of alleles and genes

- Genes are the basic units of heredity that determine the traits of an organism. They are made of DNA and are located on chromosomes.
- Alleles are different versions of the same gene that may produce different phenotypes (observable characteristics). For example, the gene for eye color has different alleles that determine whether the eyes are blue, brown, green, etc.
- Each organism inherits two alleles for each gene, one from each parent. The combination of alleles determines the genotype (genetic makeup) of the organism.
- The expression of alleles depends on their dominance and recessiveness. A dominant allele is one that masks the effect of another allele, while a recessive allele is one that is masked by another allele. For example, the allele for brown eyes is dominant over the allele for blue eyes, so an organism with one brown and one blue allele will have brown eyes.

- Some alleles are co-dominant, meaning that both alleles are expressed equally in the phenotype. For example, the alleles for blood type A and B are co-dominant, so an organism with both alleles will have blood type AB.
- Some alleles are incompletely dominant, meaning that the phenotype is a blend of both alleles. For example, the alleles for red and white flowers are incompletely dominant, so an organism with both alleles will have pink flowers.
- Some genes have multiple alleles, meaning that there are more than two possible versions of the gene. For example, the gene for blood type has three alleles: A, B, and O.
- Some traits are determined by more than one gene, meaning that they are polygenic. For example, the trait of skin color is influenced by several genes that affect the production of melanin.

Mendelian Experiments

- Mendelian experiments are the studies of inheritance patterns in pea plants conducted by Gregor Mendel, a monk and scientist, in the 19th century .
- Mendel chose pea plants because they had distinct and easily observable traits, such as flower color, seed shape, and plant height, and they could be easily cross-pollinated by transferring pollen from one plant to another .
- Mendel selected seven traits that had two contrasting forms, such as purple or white flowers, round or wrinkled seeds, and tall or dwarf plants .
- Mendel performed two types of crosses: monohybrid and dihybrid. A monohybrid cross involves one trait, such as flower color, while a dihybrid cross involves two traits, such as seed shape and color .
- Mendel observed the results of the crosses in three generations: the parental generation (P), the first filial generation (F1), and the second filial generation (F2) .
- Mendel's experiments led him to propose three laws of inheritance: the law of segregation, the law of independent assortment, and the law of dominance .
- The law of segregation states that each parent has two copies of each gene, called alleles, and that these alleles separate during gamete formation, so that each gamete receives one allele.
- The law of independent assortment states that the alleles of different genes are sorted independently of each other during gamete formation, so that the inheritance of one trait does not affect the inheritance of another trait.
- The law of dominance states that some alleles are dominant over others, meaning that they are expressed in the phenotype even if only one copy is present, while others are recessive, meaning that they are only expressed in the phenotype if two copies are present .

- Mendel's experiments and laws laid the foundation for modern genetics and the concept of alleles and genes .

Hello, I am Sydney, your AI assistant. I can help you with your study material for Remedial Biology-I KCS. Here is some information on the cell cycle for the notes of Unit 4.

Cell cycle

- The cell cycle is an ordered series of events involving cell growth and cell division that produces two new daughter cells .
- Cells on the path to cell division proceed through a series of precisely timed and carefully regulated stages of growth, DNA replication, and division that produce two genetically identical cells .
- The cell cycle can be divided into two major phases: interphase and the mitotic phase .
- Interphase is the longest phase of the cell cycle, lasting about 90% of the cycle . It consists of three subphases: G1, S, and G2 .
- G1 phase is the first gap phase, where the cell grows in size, synthesizes proteins and organelles, and prepares for DNA replication .
- S phase is the synthesis phase, where the cell replicates its DNA, resulting in two identical copies of each chromosome .
- G2 phase is the second gap phase, where the cell continues to grow and produce proteins, and checks for DNA damage and repairs any errors before entering the mitotic phase .
- The mitotic phase is the shortest phase of the cell cycle, lasting about 10% of the cycle . It consists of two stages: mitosis and cytokinesis .
- Mitosis is the process of nuclear division, where the replicated chromosomes are separated into two sets and distributed to two daughter nuclei . Mitosis can be further divided into four phases: prophase, metaphase, anaphase, and telophase .
- Cytokinesis is the process of cytoplasmic division, where the cell membrane pinches in and splits the cell into two daughter cells, each with a complete set of chromosomes and organelles .
- The cell cycle is regulated by a complex network of proteins and molecules that monitor and control the progression of the cycle .
- The main regulators of the cell cycle are the cyclins and the cyclin-dependent kinases (CDKs) . Cyclins are proteins that bind to and activate CDKs, which are enzymes that phosphorylate and activate or deactivate other proteins involved in the cell cycle .
- The levels of cyclins fluctuate throughout the cell cycle, depending on the phase and the signals received by the cell . Different cyclin-CDK complexes are responsible for triggering different events in the cell cycle, such as DNA replication, mitosis, and cell cycle arrest .
- The cell cycle also has several checkpoints, which are critical points where

the cell can pause and check for errors or signals before proceeding to the next phase . The main checkpoints are the G1/S checkpoint, the G2/M checkpoint, and the spindle checkpoint .

- The G1/S checkpoint is the point where the cell decides whether to enter the S phase and replicate its DNA or to exit the cell cycle and enter a non-dividing state called G0 . The cell checks for factors such as cell size, nutrient availability, growth factors, DNA damage, and DNA replication inhibitors .
- The G2/M checkpoint is the point where the cell decides whether to enter the mitotic phase and divide or to delay the division

Mitosis and Meiosis

Mitosis and meiosis are two types of cell division that occur in eukaryotic organisms. They have different purposes and outcomes, but they also share some similarities.

Mitosis

Mitosis is the process by which most cells in the body divide, such as skin cells, muscle cells, and nerve cells. It involves a single round of cell division, and produces two identical, diploid daughter cells. Diploid means that each cell has two sets of chromosomes, one from each parent. Mitosis is used for growth, repair, and asexual reproduction.

The stages of mitosis are:

- Prophase: The chromosomes condense and become visible. The nuclear envelope breaks down and the spindle fibers form.
- Metaphase: The chromosomes align at the middle of the cell, called the metaphase plate. The spindle fibers attach to the centromeres of the chromosomes.
- Anaphase: The sister chromatids separate and move to opposite poles of the cell. The cell elongates as the spindle fibers pull the chromosomes apart.
- Telophase: The chromosomes reach the poles and decondense. The nuclear envelope reforms and the spindle fibers disappear. Cytokinesis, the division of the cytoplasm, occurs and two daughter cells are formed.

Meiosis

Meiosis is the process by which gametes, or sex cells, are produced. It involves two rounds of cell division, and produces four haploid daughter cells. Haploid means that each cell has one set of chromosomes, half the number of the diploid cells. Meiosis is used for sexual reproduction and genetic diversity.

The stages of meiosis are:

- Meiosis I: The first round of cell division, which reduces the chromosome number from diploid to haploid.
  - Prophase I: The chromosomes condense and pair up with their homologous chromosomes, forming tetrads. Crossing over, the exchange of genetic material between homologous chromosomes, occurs and creates new combinations of genes.
  - Metaphase I: The tetrads align at the middle of the cell, with each homologous chromosome facing a different pole. The spindle fibers attach to the centromeres of the chromosomes.
  - Anaphase I: The homologous chromosomes separate and move to opposite poles of the cell. The sister chromatids remain attached.
  - Telophase I: The chromosomes reach the poles and may decondense. The nuclear envelope may reform and the spindle fibers may disappear. Cytokinesis occurs and two haploid daughter cells are formed.
- Meiosis II: The second round of cell division, which separates the sister chromatids and produces four haploid daughter cells.
  - Prophase II: The chromosomes condense and the nuclear envelope breaks down. The spindle fibers form and attach to the centromeres of the chromosomes.
  - Metaphase II: The chromosomes align at the middle of the cell, with each sister chromatid facing a different pole.
  - Anaphase II: The sister chromatids separate and move to opposite poles of the cell.
  - Telophase II: The chromosomes reach the poles and decondense. The nuclear envelope reforms and the spindle fibers disappear. Cytokinesis occurs and four haploid daughter cells are formed.

Differences and Similarities Between Mitosis and Meiosis

The main differences between mitosis and meiosis are:

- Mitosis produces two diploid daughter cells, while meiosis produces four haploid daughter cells.
- Mitosis is used for growth, repair, and asexual reproduction, while meiosis is used for sexual reproduction and genetic diversity.
- Mitosis involves one round of cell division, while meiosis involves two rounds of cell division.
- Mitosis does not involve crossing over or independent assortment, while meiosis does and creates new combinations of genes.

The main similarities between mitosis and meiosis are:

- Both are types of cell division that occur in eukaryotic organisms.
- Both involve the stages of prophase, metaphase, anaphase, and telophase, with some variations.
- Both involve the separation of chromosomes and the formation of daughter cells.

Hello, I am Sydney, your AI assistant. I can help you with your study material for Remedial Biology-I KCS. Here are some techniques to study mitosis and meiosis for the notes of the Unit 4:

Techniques to study mitosis and meiosis

- **Drawing**: Drawing both processes out on paper is a good practice exercise to deepen your understanding of mitosis and meiosis. Don't forget to label the phases, and write a brief description of what is occurring in each.
- **Simulations**: You can also use online simulations or animations as an effective review tool to review the material and to quiz yourself. You can find some examples of mitosis and meiosis simulations on Labster, Khan Academy , or other websites.
- **Experiments**: You can also perform some experiments to observe mitosis and meiosis in living cells. For example, you can prepare a squash preparation of onion root tips to observe stages of mitosis, or you can dissect a grasshopper and take out its testes to prepare a slide for the study of meiosis. You will need some equipment and materials for these experiments, such as a microscope, slides, cover-slips, stains, etc.
- **Flashcards**: You can also use flashcards to memorize the key terms and concepts related to mitosis and meiosis. You can make your own flashcards or use online flashcard apps such as Quizlet or Anki. You can also test yourself by matching the terms with the definitions or the diagrams.

Origin of life

- The origin of life is the scientific problem of how life emerged from non-living matter on Earth or elsewhere in the universe.
- There is no direct evidence or eyewitness account of the origin of life, so scientists rely on hypotheses, experiments, and observations to infer possible scenarios and mechanisms.
- One of the main challenges of the origin of life is to explain how the organic molecules that define life, such as amino acids, nucleotides, sugars, and lipids, were formed and concentrated in the prebiotic environment.
- Another challenge is to explain how these organic molecules were assembled into macromolecules, such as proteins, nucleic acids, polysaccharides, and membranes, and how these macromolecules acquired the functions of catalysis, replication, and information storage.
- A third challenge is to explain how the first self-replicating systems emerged from the macromolecules, and how they evolved into more complex and diverse forms of life.
- Several hypotheses have been proposed to address these challenges, such as the primordial soup, the iron-sulfur world, the RNA world, the lipid world, and the metabolism-first scenarios. Each hypothesis has its strengths and

weaknesses, and none of them can fully account for all the aspects of the origin of life.
- The origin of life is an interdisciplinary field that involves biology, chemistry, physics, geology, astronomy, and philosophy. It also has implications for the search for extraterrestrial life and the origin of the universe.

Evidences for Biological Evolution

Biological evolution is the change in the inherited characteristics of populations of organisms over time. Evolution explains the diversity and adaptation of life on Earth. There are many lines of evidence that support the theory of evolution, such as:

- **Fossils**: Fossils are the preserved remains or traces of organisms that lived in the past. Fossils show the sequence of appearance and disappearance of different groups of organisms over geological time. Fossils also show the transitional forms between ancestral and derived species, such as the fossils of whales, horses, and humans .
- **Comparative Anatomy**: Comparative anatomy is the study of the similarities and differences in the structures of different organisms. Comparative anatomy reveals the homologous and analogous structures of different species. Homologous structures are those that have a common origin but may have different functions, such as the forelimbs of vertebrates. Analogous structures are those that have a different origin but have similar functions, such as the wings of insects and birds. Homologous structures indicate a common ancestry, while analogous structures indicate a common adaptation .
- **Embryology**: Embryology is the study of the development of embryos of different organisms. Embryology shows that many organisms share similar stages and features in their early development, such as the presence of gill slits, tail, and pharyngeal arches in vertebrate embryos. These similarities suggest a common ancestry and a common developmental plan .
- **Molecular Biology**: Molecular biology is the study of the structure and function of molecules that are essential for life, such as DNA, RNA, and proteins. Molecular biology reveals the genetic and biochemical similarities and differences among different organisms. Molecular biology can compare the sequences of DNA, RNA, and proteins of different species and measure the degree of similarity or divergence. The more similar the sequences are, the more closely related the species are. Molecular biology can also trace the evolutionary history of genes and genomes using molecular clocks .
- **Biogeography**: Biogeography is the study of the distribution and diversity of organisms across different regions and continents. Biogeography shows that organisms that live in the same geographic area tend to be more closely related than those that live in distant areas, even if they

have different habitats. Biogeography also shows that organisms that live in isolated areas, such as islands, tend to have unique and endemic species that evolved from a common ancestor. Biogeography reflects the effects of continental drift, migration, and natural selection on the evolution of organisms .
- **Direct Observation**: Direct observation is the witnessing of evolutionary changes in populations of organisms over a short period of time. Direct observation shows that evolution can occur rapidly and visibly in response to environmental changes, such as changes in climate, food availability, predators, parasites, and human activities. Direct observation can also show the effects of artificial selection, which is the intentional breeding of organisms with desirable traits. Examples of direct observation of evolution include the evolution of antibiotic resistance in bacteria, pesticide resistance in insects, beak size variation in finches, and domestication of animals and plants .

These evidences for biological evolution provide a strong and consistent support for the theory of evolution and its mechanisms. They also help us understand the origin, diversity, and adaptation of life on Earth.

Mechanism of evolution

- Evolution is the process by which modern organisms have descended from ancient ancestors.
- Evolution is responsible for both the remarkable similarities and the amazing diversity of life on Earth.
- Evolution occurs when the frequency of alleles (variants of genes) in a population changes over generations.
- There are several mechanisms that can cause evolution, such as natural selection, mutation, genetic drift, migration and non-random mating .
- These mechanisms can affect the genetic variation and the adaptation of populations in different ways.

Natural selection

- Natural selection is the process by which individuals with certain traits have higher survival and reproduction rates than others in a given environment .
- Natural selection can increase the frequency of beneficial alleles and decrease the frequency of harmful alleles in a population .
- Natural selection can also result in different types of adaptations, such as directional selection, stabilizing selection, disruptive selection, sexual selection and artificial selection .
- Examples of natural selection include the evolution of antibiotic resistance in bacteria, the evolution of camouflage in animals, the evolution of beak shapes in finches and the evolution of human skin color .

Mutation

- Mutation is the process by which random changes occur in the DNA sequence of an organism .
- Mutation can introduce new alleles into a population or change the frequency of existing alleles .
- Mutation can be beneficial, harmful or neutral depending on the effect on the phenotype and the fitness of the organism .
- Mutation is the ultimate source of genetic variation and the raw material for evolution .
- Examples of mutation include the evolution of sickle cell anemia, the evolution of lactose tolerance, the evolution of color blindness and the evolution of hemophilia .

Genetic drift

- Genetic drift is the process by which random fluctuations in allele frequencies occur due to chance events in small populations .
- Genetic drift can cause alleles to become fixed or lost in a population, reducing genetic variation .
- Genetic drift can also cause populations to diverge genetically, increasing genetic differentiation .
- Genetic drift can be influenced by factors such as population size, population bottlenecks and founder effects .
- Examples of genetic drift include the evolution of coat color in mice, the evolution of blood types in humans, the evolution of island dwarfism and the evolution of genetic diseases .

Migration

- Migration is the process by which individuals move between populations, carrying their alleles with them .
- Migration can increase or decrease the frequency of alleles in a population, depending on the direction and magnitude of gene flow .
- Migration can also reduce genetic differences between populations, increasing genetic homogeneity .
- Migration can be influenced by factors such as geographic barriers, environmental gradients, dispersal abilities and mating preferences .
- Examples of migration include the evolution of cichlid fish, the evolution of human populations, the evolution of insecticide resistance and the evolution of hybrid zones .

Non-random mating

- Non-random mating is the process by which individuals choose their mates based

Variation

Variation is the difference between individuals of the same species, caused by genetic and environmental factors . Variation can be classified into two types: continuous and discontinuous.

Continuous variation

Continuous variation is when the data come in a range of values, such as height, weight, or skin color. Continuous variation is influenced by both genes and the environment. For example, a person's height is determined by the genes inherited from their parents, but also by their nutrition, health, and age. Continuous variation can be represented by a line graph or a histogram.

Discontinuous variation

Discontinuous variation is when the data come in distinct groups, such as blood type, eye color, or gender. Discontinuous variation is usually determined by a single gene or a few genes, and is not affected by the environment. For example, a person's blood type is determined by the alleles they inherit from their parents, and cannot be changed by their diet, lifestyle, or climate. Discontinuous variation can be represented by a bar chart or a pie chart.

Sources of variation

Variation can arise from two main sources: mutation and recombination .

Mutation

Mutation is a change in the DNA sequence of a gene or a chromosome. Mutation can occur randomly due to errors in DNA replication, or due to exposure to radiation, chemicals, or viruses. Mutation can create new alleles or alter the expression of existing alleles. Mutation is the ultimate source of genetic variation, as it introduces new genetic information into a population .

Recombination

Recombination is the exchange of genetic material between homologous chromosomes during meiosis. Recombination can create new combinations of alleles on the same chromosome, or shuffle the alleles between different chromosomes. Recombination increases the genetic diversity of gametes, and thus the offspring. Recombination is influenced by the number and location of crossing-over events, and the independent assortment of chromosomes .

Importance of variation

Variation is important for the survival and evolution of a species, as it provides the raw material for natural selection. Natural selection is the process by which individuals with certain traits have a higher chance of surviving and reproducing than others in a given environment. Natural selection can result in adaptation, speciation, or extinction, depending on the environmental conditions and the degree of variation in the population .

Types of natural selection

Natural selection can act on variation in different ways, depending on the distribution and fitness of the traits in the population. There are three main types of natural selection: directional, stabilizing, and disruptive .

Directional selection

Directional selection is when one extreme of a trait is favored over the other, and the average value of the trait shifts in that direction. Directional selection can occur when the environment changes, or when a new environment is colonized. For example, directional selection can favor darker fur color in mice that live in a dark soil, or longer beaks in birds that feed on larger seeds .

Stabilizing selection

Stabilizing selection is when the intermediate value of a trait is favored over the extremes, and the average value of the trait remains the same. Stabilizing selection can occur when the environment is stable, or when the optimal value of the trait is well-adapted. For example, stabilizing selection can favor medium body size in lizards that need to balance heat loss and heat gain, or medium birth weight in humans that need to balance survival and delivery .

Disruptive selection

Disruptive selection is when both extremes of a trait are favored over the intermediate, and the average value of the trait diverges in both directions. Disruptive selection can occur when the environment is heterogeneous, or when there is competition for resources. For example, disruptive selection can favor large and small seeds in plants that face different predation pressures, or large and small beaks in birds that feed on different types of seeds .

Examples of variation and natural selection

Some examples of variation and natural selection in nature are:

- The peppered moth (Biston betularia) is a classic example of directional selection. The peppered moth has two color variants: light and dark.

Before the Industrial Revolution, the light variant was more common, as it blended in with the lichen-covered trees. However,

Mutation and Recombination

- Mutation and recombination are two mechanisms that alter the DNA sequence of a genome.
- Mutation is an alteration in a nucleotide sequence while recombination alters a large region of the genome.
- Mutation is the original source of all genetic variation, but its rate is usually low and its impact on allele frequencies is usually not large.
- Recombination is the process of exchanging genetic material between different DNA molecules, resulting in new combinations of alleles.
- Recombination can occur throughout the genome of diploid organisms, using one or a small number of common enzymatic pathways.
- Recombination can increase genetic diversity and facilitate adaptation by breaking negative associations among selected alleles.
- Mutation and recombination are both important for evolution, but recombination is considered as the major driving force of evolution because it affects a larger portion of the genome and creates more variation.